

Companion Document: update_tau.R

q() — Closed-Form MLE for Noise Precisions

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0.1 Overview

update_tau.R implements the variational update for $\tau = (\tau_1, \dots, \tau_p)$, the per-feature noise precisions in the model

$$Y = LF^\top + E, \quad E_{ij} \sim \mathcal{N}(0, \tau_j^{-1})$$

This module is **unique among the four update modules** in two important ways:

1. **No EBNM dependency** — the update is a closed-form maximum likelihood estimate, not an empirical Bayes normal means problem. The module uses pure base R with no external packages.
2. **No per-k loop** — all K factors are handled simultaneously through matrix operations. The entire update is computed in a handful of matrix multiplications.

The module exports three functions:

Function	Returns	Purpose
compute_var_term	$n \times p$ matrix	Posterior variance correction term
compute_expected_residual_sqn	$n \times p$ matrix	Expected squared residual under q
update_tau	list (Tau, R2_bar, Var_Term, elbo_proxy)	Full update + ELBO proxy

0.2 Mathematical Background

0.2.1 The CAVI Objective for τ_j

Taking the variational expectation of the complete-data log-likelihood with respect to $q(L)$ and $q(F)$, the optimal $q(\tau_j)$ is Gamma-distributed. Under a flat (improper) prior on τ_j , the CAVI fixed-point reduces to a column-wise MLE:

$$\tau_j = \frac{n}{\sum_i \bar{R}_{ij}^2}$$

where $\bar{R}_{ij}^2$ is the **expected squared residual** under the current posterior $q(L) \quad q(F)$.

0.2.2 Expected Squared Residual

The expected squared residual cannot be computed naively as $(Y - \bar{L}\bar{F}^\top)^2$. Posterior uncertainty in L and F inflates the true expectation:

$$\bar{R}_{ij}^2 = \mathbb{E}_q \left[\left(Y_{ij} - \sum_k L_{ik} F_{jk} \right)^2 \right] = \left(Y_{ij} - \sum_k \bar{L}_{ik} \bar{F}_{jk} \right)^2 + \text{VT}_{ij}$$

where VT_{ij} is the variance correction term.

The second term, VT , accounts for the fact that $\mathbb{E}[(LF^\top)^2] \neq (\bar{L}\bar{F}^\top)^2$ when L and F have nonzero posterior variance.

0.2.3 Variance Correction (Var_Term)

Under the mean-field factorization $q(L) \ q(F)$ with independent factors:

$$\begin{aligned} \text{VT}_{ij} &= \sum_k \mathbb{E}[L_{ik}^2] \mathbb{E}[F_{jk}^2] - \sum_k \bar{L}_{ik}^2 \bar{F}_{jk}^2 \\ &= (\mathbb{E}[L^2] \mathbb{E}[F^2]^\top)_{ij} - (\bar{L}^2 (\bar{F}^2)^\top)_{ij} \end{aligned}$$

where $\mathbb{E}[L_{ik}^2]$ and $\mathbb{E}[F_{jk}^2]$ are posterior second moments (not squared means).

This simplification uses independence across factors and across the $q(L) \ q(F)$ factorization. By Jensen's inequality, $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ for any random variable X , so $\text{VT}_{ij} \geq 0$ for all i, j .

VT is exactly zero only when **both** $\mathbb{E}[L^2] = \bar{L}^2$ **and** $\mathbb{E}[F^2] = \bar{F}^2$ — i.e., all posterior variances are identically zero (a degenerate point-mass posterior).

0.2.4 Why Ignoring the Correction Is Dangerous

If one naively substitutes $\mathbb{E}[L^2] \rightarrow \bar{L}^2$ and $\mathbb{E}[F^2] \rightarrow \bar{F}^2$ (ignoring posterior uncertainty), the effective $\bar{R}_{ij}^2$ is **underestimated**, causing τ_j to be **overestimated** (noise underestimated). In practice this bias is 3–5 $\times$ in typical simulations (see Demo 3).

0.2.5 ELBO Proxy

The module also computes an ELBO proxy (the τ -dependent terms of the evidence lower bound):

$$\text{ELBO}_{\text{proxy}} = \sum_j \left[\frac{n}{2} \log(\tau_j) - \frac{\tau_j}{2} \sum_i \bar{R}_{ij}^2 \right]$$

At the MLE solution $\tau_j = n / \sum_i \bar{R}_{ij}^2$, this is maximized over τ . Monitoring $\text{ELBO}_{\text{proxy}}$ across iterations provides a convergence diagnostic: it should be non-decreasing. A decrease signals a bug in the implementation.

0.3 Function Reference

0.3.1 compute_var_term

```
compute_var_term(EL, EL2, EF, EF2)
```

Arguments

Argument	Shape	Description
EL	$n \times K$	Posterior means $E[L]$
EL2	$n \times K$	Posterior second moments $E[L^2]$
EF	$p \times K$	Posterior means $E[F]$
EF2	$p \times K$	Posterior second moments $E[F^2]$

Returns an $n \times p$ matrix.

Formula

```
(EL2 %*% t(EF2)) - (EL^2 %*% t(EF^2))
```

Notes - All entries are ≥ 0 by Jensen's inequality. - Entries are zero when the corresponding column of EL (or EF) has zero posterior variance, i.e., $EL2[:,k] = EL[:,k]^2$. - This function is a pure computation — no side effects, no floor applied.

0.3.2 compute_expected_residual_sq

```
compute_expected_residual_sq(Y, EL, EL2, EF, EF2)
```

Arguments

Argument	Shape	Description
Y	$n \times p$	Observed genomics matrix
EL	$n \times K$	Posterior means $E[L]$
EL2	$n \times K$	Posterior second moments $E[L^2]$
EF	$p \times K$	Posterior means $E[F]$
EF2	$p \times K$	Posterior second moments $E[F^2]$

Returns an $n \times p$ matrix $\bar{R}^2$.

Formula

```
(Y - EL %*% t(EF))^2 + compute_var_term(EL, EL2, EF, EF2)
```

Notes - The first term is the squared residual at posterior means (what naive MLE would use). - The second term inflates it to account for posterior uncertainty. - All entries are 0 because both summands are non-negative.

0.3.3 update_tau

```
update_tau(Y, EL, EL2, EF, EF2, tau_floor = 1e-8)
```

Arguments

Argument	Shape	Default	Description
Y	$n \times p$	—	Observed genomics matrix
EL	$n \times K$	—	Posterior means $E[L]$
EL2	$n \times K$	—	Posterior second moments $E[L^2]$
EF	$p \times K$	—	Posterior means $E[F]$
EF2	$p \times K$	—	Posterior second moments $E[F^2]$
tau_floor	scalar	1e-8	Minimum τ value (prevents Tau = Inf)

Returns a named list:

Element	Shape	Description
Tau	p-vector	Estimated noise precisions $\tau_1, \dots, \tau_p$
R2_bar	$n \times p$	Expected squared residuals
Var_Term	$n \times p$	Variance correction term
elbo_proxy	scalar	ELBO proxy (-dependent terms)

Algorithm

```
Var_Term <- compute_var_term(EL, EL2, EF, EF2)
R2_bar   <- compute_expected_residual_sq(Y, EL, EL2, EF, EF2)
RSS_j    <- colSums(R2_bar)                                # length-p vector
Tau      <- n / pmax(RSS_j, n * tau_floor)                 # MLE with floor
elbo_proxy <- sum(n/2 * log(Tau) - Tau/2 * RSS_j)
```

Numerical stability

`pmax(RSS_j, n * tau_floor)` prevents division by near-zero RSS when the reconstruction is nearly perfect. Without this floor, $\tau_j \rightarrow \infty$ (infinite precision), which can destabilize downstream L and F updates. The floor corresponds to a maximum τ of $1/\text{tau_floor} = 1e8$.

No per-k loop

Unlike `update_beta_all`, `update_L_all`, and `update_F_all`, this function processes all K factors in a single pair of matrix multiplications. The sum over k in the `Var_Term` formula collapses into matrix products.

0.4 Key Properties

Property	Statement
Update type	Closed-form MLE (NOT EBNM)
Loop structure	No per-k loop; fully vectorized
External dependencies	None — pure base R
Output positivity	$\text{Tau} > 0$ guaranteed by <code>tau_floor</code>
VT sign	≥ 0 everywhere (Jensen's inequality)
ELBO monotonicity	$\text{ELBO}_{\text{proxy}}$ non-decreasing at convergence
Perfect reconstruction	Tau hits ceiling $1/\text{tau_floor}$ when $\text{RSS} = 0$
Heteroscedastic noise	Each τ_j estimated independently per column

0.5 Test Suite

File: `tests/test_update_tau.R` **Total:** 27 tests across 9 groups (all passing)

0.5.1 T1: Mathematical Identities (5 tests)

Verify that each formula is implemented exactly as derived.

Test	Assertion
T1.1	<code>Var_Term == EL2 %*% t(EF2) - EL^2 %*% t(EF^2)</code> (to machine precision)
T1.2	<code>all(Var_Term >= 0)</code>
T1.3	<code>R2_bar == (Y - EL %*% t(EF))^2 + Var_Term</code>
T1.4	<code>Tau == n / colSums(R2_bar)</code> (when above floor)

Test	Assertion
T1.5	ELBO proxy formula matches manual computation

0.5.2 T2: Perfect Reconstruction (3 tests)

Edge case where Y is exactly reproduced by posterior means and posteriors collapse to point masses.

Test	Assertion
T2.1	$\text{Var_Term} = 0$ when $\text{EL2} = \text{EL}^2$ and $\text{EF2} = \text{EF}^2$
T2.2	$\text{R2_bar} = 0$ when $Y = \text{EL} \cdot t(\text{EF})$ and no posterior variance
T2.3	Tau hits ceiling ($1/\text{tau_floor}$) for zero-residual case

0.5.3 T3: Known Noise Recovery (3 tests)

Simulates $Y = LF^\top + \mathcal{N}(0, \tau_j^{-1})$ with zero posterior variance; checks that estimated τ recovers truth.

Test	Assertion
T3.1	$\text{cor}(\text{Tau_est}, \text{Tau_true}) \approx 0.9$ ($n = 200$)
T3.2	Heteroscedastic columns separated correctly
T3.3	Larger n gives better recovery (consistency check)

0.5.4 T4: Variance Correction (3 tests)

Demonstrates that the Var_Term matters quantitatively.

Test	Assertion
T4.1	Var_Term inflates R2_bar relative to naive $(Y - \text{pred})^2$
T4.2	Ignoring correction overestimates $\text{mean}(\text{Tau})$
T4.3	Larger posterior variance $\rightarrow$ larger Var_Term

0.5.5 T5: ELBO Proxy (3 tests)

Test	Assertion
T5.1	elbo_proxy matches manual formula
T5.2	elbo_proxy is finite (no NaN/Inf)
T5.3	MLE Tau maximizes elbo_proxy ($\text{ELBO_MLE} = \text{ELBO_perturbed}$)

0.5.6 T6: Dimension & Shape (2 tests)

Test	Assertion
T6.1	<code>dim(Var_Term) == c(n, p)</code>
T6.2	<code>length(Tau) == p</code>

0.5.7 T7: Numerical Stability (4 tests)

Test	Assertion
T7.1	<code>Y = 0</code> produces finite, positive Tau
T7.2	<code>1×1×1</code> case (<code>n=1</code> , <code>p=1</code> , <code>K=1</code>) does not error
T7.3	Large entries (<code>1e6</code> scale) do not overflow
T7.4	<code>tau_floor</code> prevents <code>Tau = Inf</code>

0.5.8 T8: Pipeline Interaction (2 tests)

Test	Assertion
T8.1	Tau is positive, finite, length p (valid downstream input)
T8.2	Tau can be used in <code>A_L</code> computation without error

0.5.9 T9: V2.R Consistency (2 tests)

Compares output against the corresponding lines in `code/Supervised_Bayesian_MF_V2.R`.

Test	Assertion
T9.1	<code>Var_Term</code> and <code>Tau</code> match V2.R lines 374–376 to within <code>1e-12</code>
T9.2	<code>elbo_proxy</code> matches V2.R line 379 to within <code>1e-12</code>

0.6 Demos

File: `demos/demo_update_tau.R`

Each demo is a narrative scenario with printed output. There are no PASS/FAIL assertions — these are exploratory illustrations, not unit tests.

0.6.1 Demo 1: Anatomy of Var_Term and R2_bar

Purpose: Decompose $R2_bar$ into its two additive components and show that Var_Term is non-negligible.

Setup: Small $n=20$, $p=10$, $K=3$ example with realistic posterior variances.

Key output: Mean ratio $VT_{ij}/\bar{R}_{ij}^2$; illustrates that ignoring the correction would systematically bias τ upward.

0.6.2 Demo 2: Known Noise Recovery

Purpose: Validate that `update_tau` recovers true per-feature precisions under controlled conditions.

Setup: $Y = LF^\top + E$ with $\tau_j \sim \text{Uniform}(0.5, 5)$; point-mass posteriors ($EL2 = EL^2$, $EF2 = EF^2$); $n=200$.

Key output: $\text{cor}(\text{estimated } \tau, \text{true } \tau) \approx 0.9+$; scatter plot of estimated vs. true.

0.6.3 Demo 3: Variance Correction Matters

Purpose: Quantify the bias introduced by ignoring $EL2$ and $EF2$.

Setup: Same Y as Demo 2, but run `update_tau` twice — once with true second moments, once substituting $EL2 \leftarrow EL^2$ and $EF2 \leftarrow EF^2$.

Key output: $\text{mean}(\tau_{\text{naive}}) / \text{mean}(\tau_{\text{correct}}) \approx 3\text{--}5\times$; shows systematic overestimation of precision (underestimation of noise) when the correction is omitted.

0.6.4 Demo 4: Heteroscedastic Noise

Purpose: Confirm that the column-wise MLE correctly separates features with different noise levels.

Setup: $p=150$ features split into two groups — high-precision ($\tau = 5$, low noise) and low-precision ($\tau = 0.5$, high noise).

Key output: Estimated τ clearly bimodal; group means close to 5 and 0.5 respectively.

0.6.5 Demo 5: ELBO as Convergence Monitor

Purpose: Show that `elbo_proxy` is a valid convergence diagnostic that increases (or plateaus) across iterations.

Setup: 3 manual CAVI iterations — `update L` → `update F` → `update` — on a small example.

Key output: $ELBO_{\text{proxy}}$ at each iteration; should be non-decreasing. A decrease would indicate a bug in `L` or `F` update.

0.7 Pipeline Context

0.7.1 Data Flow

```
update_L_all → EL, EL2
                                     → update_tau → Tau (p-vector)
update_F_all → EF, EF2
```

Upstream: `update_tau` consumes posterior moments from `update_L.R` and `update_F.R`. Specifically, it requires both the posterior means (EL, EF) and the posterior second moments (EL2, EF2).

Downstream: The output `Tau` feeds back into: - `update_L_k` via the `A_L` precision accumulation: `A_L <- A_L + Tau[j] * EF2[j, k]` - `update_F_k` via the `A_F` precision accumulation: `A_F <- A_F + Tau[j] * EL2[i, k]`

0.7.2 Position in CAVI Loop

```
Repeat until convergence:
  1. update_beta_all(...)      # q() update - scalar EBNM per factor
  2. update_L_all(...)         # q(L) update - vector EBNM, dual-source
  3. update_F_all(...)         # q(F) update - vector EBNM, genomics only
  4. update_tau(...)           # q() update - closed-form MLE <-- THIS MODULE
  5. Check convergence ( $\Delta L + \Delta < \text{tol}$ , ELBO non-decreasing)
```

0.7.3 V2.R Reference Lines

The logic in this module corresponds to the following lines in `code/Supervised_Bayesian_MF_V2.R`:

V2.R Lines	Content
374–376	<code>Var_Term, R2_bar, Tau</code> computation
379	ELBO proxy accumulation

Tests T9.1 and T9.2 verify exact agreement (to 1e-12) between this module and those V2.R lines.

File	Role
------	------

0.8 Related Files

File	Role
code/update_tau.R	Source implementation
tests/test_update_tau.R	27-test suite (9 groups)
demos/demo_update_tau.R	5 narrative demo scenarios
derivations/qTau/qTau_update_derivation.tex	Full mathematical derivation (variance correction proof)
code/Supervised_Bayesian_MF_V2.R	Main algorithm (lines 374–376, 379)
code/update_L.R	Upstream: produces EL, EL2
code/update_F.R	Upstream: produces EF, EF2
code/SupervisedMF_Context.md	AI/code quick-reference for the full R implementation
PROJECT_STATUS.md	Project-level documentation and session log

0.9 Quick Reference Card

```
# Minimal usage
source("code/update_L.R")    # provides EL, EL2
source("code/update_F.R")    # provides EF, EF2
source("code/update_tau.R")

result    <- update_tau(Y, EL, EL2, EF, EF2)
Tau       <- result$Tau      # p-vector of noise precisions
elbo_proxy <- result$elbo_proxy # convergence monitor

# Internals
Var_Term <- compute_var_term(EL, EL2, EF, EF2)          # n × p, >= 0
R2_bar   <- compute_expected_residual_sq(Y, EL, EL2, EF, EF2) # n × p

# MLE formula (closed form, no EBNM, no per-k loop)
Tau[j]   <- n / sum(R2_bar[, j])

# ELBO proxy
elbo_proxy <- sum(n/2 * log(Tau) - Tau/2 * colSums(R2_bar))
```

Critical reminders - VT uses **second moments** $\mathbb{E}[L^2]$, $\mathbb{E}[F^2]$ — not $\bar{L}^2$, $\bar{F}^2$. Confusing these is the most common implementation error and causes 3–5× overestimation of τ . - tau_floor is applied to RSS (not to Tau directly): `pmax(RSS_j, n * tau_floor)`. - ELBO_{proxy} should be **non-decreasing** across

CAVI iterations. If it drops, check the L or F update first. - This is the **only module** with no EBNM call and no `ebnm` package dependency.